

Full Stack Development with MERN

Project Documentation

1. Introduction

- Project Title: Import Data using Transform Maps
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This project demonstrates importing structured employee data from an external spreadsheet into ServiceNow using Import Sets and Transform Maps.

2. Project Overview

- Purpose: To provide a controlled workflow for loading bulk employee data from a spreadsheet into ServiceNow, transforming it into target records, validating the results, preventing duplicates, and presenting the data through reports and an analytics dashboard.
- Features: Spreadsheet preparation; Employee Test target table; Employee Import staging table; Transform Map; Auto Map Matching Fields; Mapping Assist; transformation and validation; Coalesce-based duplicate prevention; repeated imports; Transform History; department, location and employee-list reports; Employee Analytics Dashboard.

3. Architecture

- Frontend: Not specified in the source. The source describes ServiceNow forms, reports, and dashboards rather than a separate React frontend.
- Backend / Application Platform: ServiceNow Import Sets, Load Data, Transform Maps, Mapping Assist, Auto Map Matching Fields, Transform Engine, Transform History, reports and dashboards.
- Database / Data Tables: Employee Import staging table (u_employee_import) receives source spreadsheet data. Employee Test target table (u_employee_test) stores transformed employee records. The source does not specify MongoDB or another external database.

4. Setup Instructions

- Prerequisites: ServiceNow environment with the required permissions and access to the sample employee spreadsheet.
- Installation / Configuration: Prepare the spreadsheet; create the Employee Test table and fields; create the Employee Import Import Set table; load the spreadsheet; create the Sample Spreadsheet Import Transform Map; map source fields to target fields; save and run the transform.
- Environment Variables: Not specified in the source document.

5. Folder Structure

- Client: Not applicable / not specified because the project is implemented through ServiceNow configuration.
- Server: Not applicable as a separate Node.js server; ServiceNow provides the application environment, Import Set processing and transformation workflow.

6. Running the Application

- Prepare and load the employee spreadsheet through ServiceNow Load Data.
- Select the existing Employee Import table for subsequent spreadsheet imports.
- Run Transform to process the staged rows into Employee Test.
- Use Transform History to inspect inserted, updated, or ignored rows.
- Use the Employee Analytics Dashboard to view the resulting reports.

7. API Documentation

- No external REST/API endpoints are documented in the supplied source.
- The documented integration mechanism is ServiceNow Import Sets and Transform Maps for spreadsheet-to-table data movement.

8. Authentication

- Authentication and authorization implementation details are not specified in the source document.
- The source mentions that dashboards can be restricted to users, groups, or roles, but it does not define a separate authentication/token/session design.

9. User Interface

- ServiceNow interfaces shown in the source include the Employee Import table, Transform Map configuration, Employee Test records, Transform History, report configuration, and Employee Analytics Dashboard.

Representative screenshots are taken from the supplied source document to demonstrate the documented workflow.

Import Data using Transform Maps (Spreadsheet)

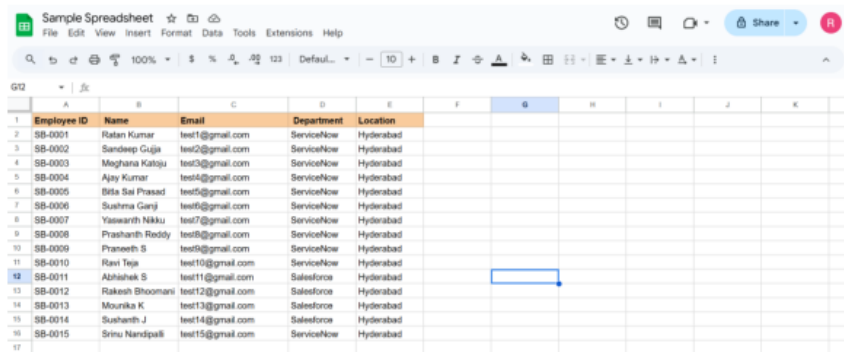
Description:

This micro project focuses on importing structured data from an external spreadsheet into the **ServiceNow** platform using Import Sets and Transform Maps. The project simulates a real-world scenario where bulk employee data is received in Excel format and must be migrated into ServiceNow efficiently and accurately. The spreadsheet data is first loaded into an Import Set table, which serves as a staging area, and a Transform Map is then configured to map source fields to the appropriate target table fields. Upon executing the transformation, records are created in the target table and validated to ensure data integrity. This project helps beginners understand the end-to-end data import process in ServiceNow, including data loading, field mapping, transformation execution, and result verification.

Step by Step process:

Phase 1: Prepare the sheets

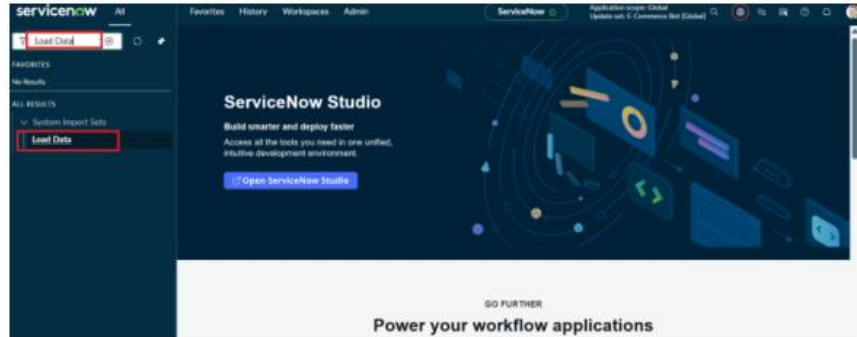
1. Open New Google Spreadsheet
2. Create New Google Spreadsheet with sample data as shown below



Employee ID	Name	Email	Department	Location
SB-0001	Ratan Kumar	test1@gmail.com	ServiceNow	Hyderabad
SB-0002	Sandeep Guja	test2@gmail.com	ServiceNow	Hyderabad
SB-0003	Meghana Katoju	test3@gmail.com	ServiceNow	Hyderabad
SB-0004	Ajay Kumar	test4@gmail.com	ServiceNow	Hyderabad
SB-0005	Bitu Sai Prasad	test5@gmail.com	ServiceNow	Hyderabad
SB-0006	Suchma Ganj	test6@gmail.com	ServiceNow	Hyderabad
SB-0007	Yashwanth Nikku	test7@gmail.com	ServiceNow	Hyderabad
SB-0008	Prashanth Reddy	test8@gmail.com	ServiceNow	Hyderabad
SB-0009	Praneeth S	test9@gmail.com	ServiceNow	Hyderabad
SB-0010	Ravi Teja	test10@gmail.com	ServiceNow	Hyderabad
SB-0011	Abhishek S	test11@gmail.com	Salesforce	Hyderabad
SB-0012	Rakesh Bhoomani	test12@gmail.com	Salesforce	Hyderabad
SB-0013	Mounika K	test13@gmail.com	Salesforce	Hyderabad
SB-0014	Sushanth J	test14@gmail.com	Salesforce	Hyderabad
SB-0015	Srinu Nandipati	test15@gmail.com	ServiceNow	Hyderabad

3. Download the spreadsheet as Sample Spreadsheet.xlsx format.

Figure 1. Source workflow screenshot — source page 1.



Click on Load Data and create an import table
Fill details as shown below

- Label: Employee Import
- Name: u_employee_import (auto populated)

- Click on Submit

Once we click on Submit, we can see the success message as shown below.

Also we need to make sure to click on Create Transform Map.

Figure 2. Source workflow screenshot — source page 5.



Phase 4: Create Transform Map

In this phase we need to create the Transform Map by clicking on the Create Transform Map button as shown in above screenshot.

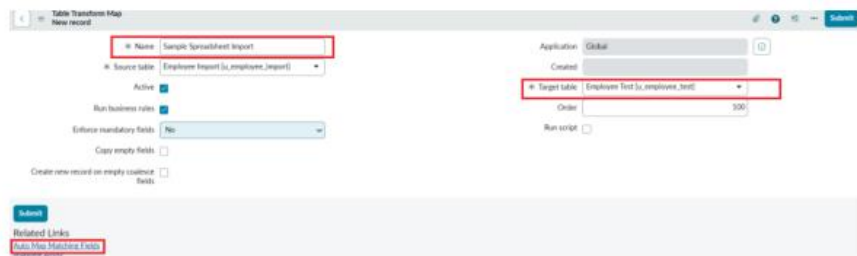
Give inputs as shown below.

Name: Sample Spreadsheet Import

Source Table: Employee Import is auto-populated

Target Table: Employee Test - as we created earlier

Click on Auto Map Matching fields.



Now click on Mapping Assist and map the fields with Source & Target Table



Click on Save

Make sure the both Source & Target table fields are mapped as shown below.

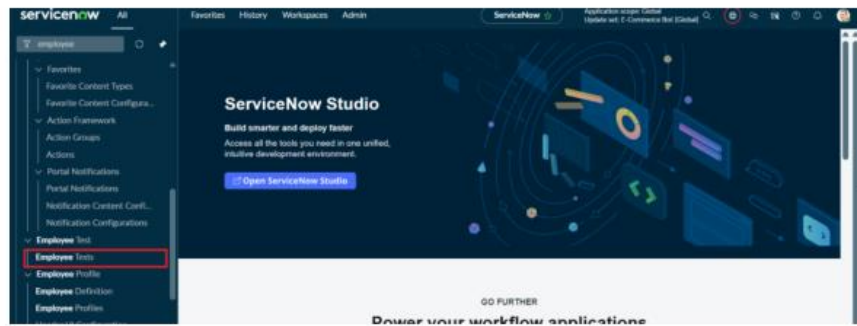
Figure 3. Source workflow screenshot — source page 6.

Progress	
Name	Transforming: RET0000001
State	Complete
Completion code	Success
Message	Initialization complete
Next steps...	
RET0000001	Go to the import sets for this data load
Transform history	Show the transform history, related errors and log
Import log	View the import log

Phase 5: Transform Data & Validate

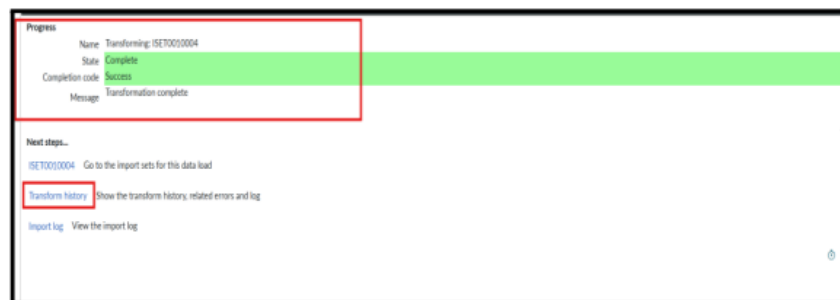
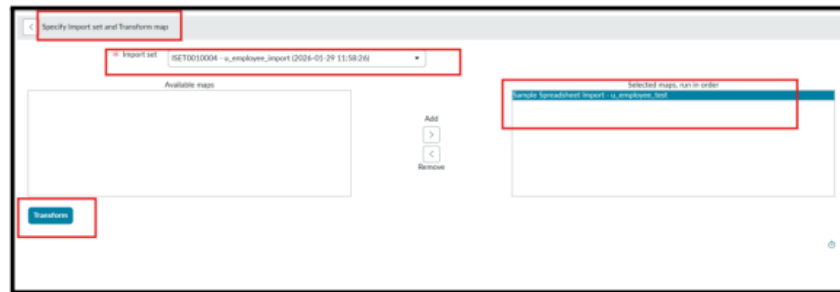
Navigation:

- Open Employee Test Table from application navigator as shown below.



- As our import got succeeded, all the records are created in the Employee Test Table.

Figure 4. Source workflow screenshot — source page 8.



- Now the State is completed and Success.
- Now again click on the Transform history then it shows the newly added data and its status of Inserted and Skipped values out of the given data.

Started	State	Completed	Run time	Set	Import set table	Total	Inserts	Updates	Ignored	Skipped	Errors	Transform Map
2026-01-29 03:47:37	Complete	2026-01-29 03:47:38	1 Second	ISET0000000	Employee Import (u_employee_import)	4	2	2	0	0	0	Sample Spreadsheet Import

Figure 5. Source workflow screenshot — source page 15.

PHASE 8 : Create Reports on Employee Test Table

Navigation Steps :

- Open All menus, Search for **Reports**.
- Under **Usage and governance** Select Reports.
- Click the **New** button to create a new report.
- Create **3 Reports** as follows.

Report 1: Employees by Department

- In Data
- Report name : **Employees by Department**.
- Source Type : **Table**.
- Table : **Employee Test**, Now click the Next button.

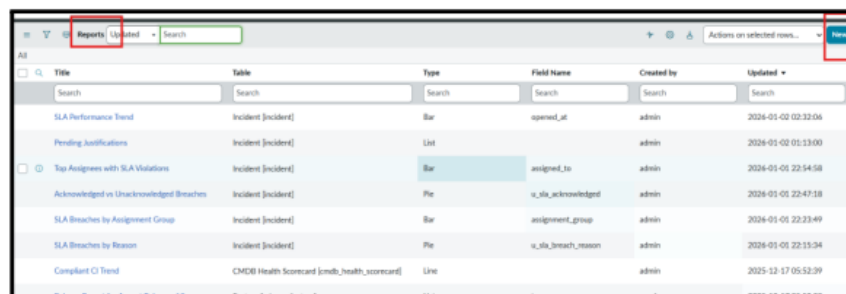
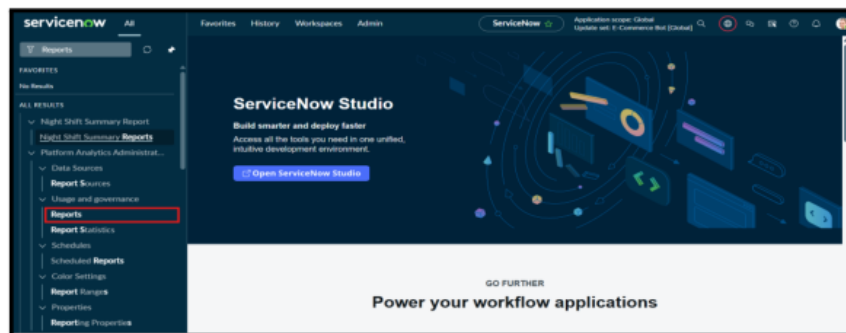
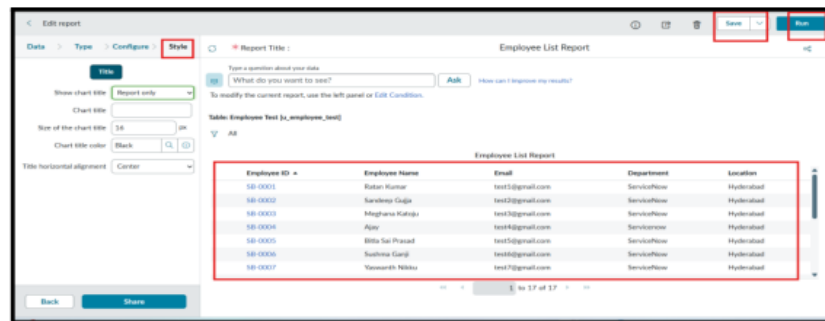
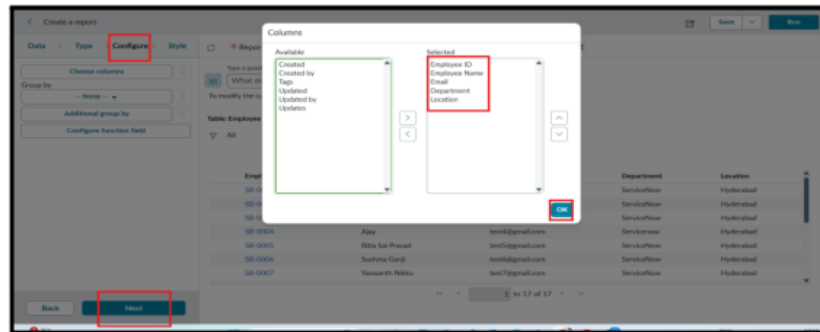


Figure 6. Source workflow screenshot — source page 17.



PHASE 9 : Add Reports to Dashboard

Create Dashboard

Navigation :

- Open All menus → Search for Dashboards.
- Under Self Service → Click Dashboards.
- On the top right corner Click Create Dashboards or /
- You can create Directly by using the backend name [PA_DASHBOARDS.FORM].
- Then Click New to Create new Dashboards.

Figure 7. Source workflow screenshot — source page 24.



- Here the three Reports are added successfully by varying **Department , Location and List Report.**
- We can also restrict the **Users,Groups and Roles** From Share Option.

Conclusion:

1. This project successfully implemented an automated employee data management solution using ServiceNow Import Sets and Transform Maps, ensuring accuracy, efficiency, and data integrity throughout the employee onboarding and update process.
2. A key enhancement in this project was the use of Coalesce in the Transform Map. By enabling Coalesce on the Employee ID field, the system intelligently identifies existing employee records during every import. This prevents the creation of duplicate records and ensures that repeated imports update existing data instead of inserting redundant entries. As a result, the employee database remains clean, reliable, and consistent, which is critical in real-world enterprise environments where data may be imported multiple times from external systems.
3. In addition to data integrity, the project was extended with Reports and Dashboards to provide meaningful insights and real-time visibility into employee data and import activities. Custom reports were created to track:

Figure 8. Source workflow screenshot — source page 30.

10. Testing

- Functional testing: Validate that transformed records appear in Employee Test.
- Import validation: Use Transform History to inspect inserted, updated, and ignored rows.
- Duplicate prevention: Enable Coalesce so repeated imports update the same employee records rather than creating duplicate records.
- Repeated import scenario: The source demonstrates new rows being inserted, existing rows being updated when values change, and repeated identical rows being ignored.

- Quantitative performance metrics such as latency, throughput, CPU usage, or memory usage are not specified in the source.

11. Screenshots or Demo

- The supplied source document contains screenshots of the import, transformation, validation, reporting, and dashboard workflow.
- No external demo URL is specified in the source.

12. Known Issues

- The workflow depends on correctly prepared spreadsheet data and correctly configured source-to-target mappings.
- The source does not document automated API integration, production deployment details, quantified performance testing, or a dedicated external authentication design.
- No other specific software bugs are documented in the source.

13. Future Enhancements

- Automate scheduled or recurring imports from additional external data sources.
- Add more data-quality validation rules before transformation.
- Expand dashboard and report metrics for employee and import monitoring.
- Introduce formal performance and security testing for production use.
- These are proposed extensions; they are not described as implemented features in the source.

Source basis: Import Data using Transform Maps (Spreadsheet). All project-specific details above are restricted to information supported by that source; unsupported template fields are explicitly marked as not specified.